MemoryRecall — Gamified Active Recall App

Full-Stack Architecture & Implementation Powered by Sarvam AI

1. Executive Summary & Project Description

MemoryRecall is an interactive, gamified active recall and spaced repetition learning application. It empowers students, educators, and lifelong learners to transform raw learning materials—including **handwritten notes, textbook photos, screenshots, and typed summaries**—into durable, long-term memory. By integrating **Sarvam AI's Document Intelligence (Doc AI)** for handwriting OCR and the **sarvam-105b-conversations** large language model for pedagogical study material synthesis, the application creates a continuous learning loop: **Capture** → **Extract** → **Understand** → **Generate** → **Practice** → **Review** → **Retain**.

> ***Core Value Proposition:*** *Most flashcard apps require tedious manual data entry. MemoryRecall automates transcription using cutting-edge handwriting OCR, leverages AI to generate conceptual recall prompts rather than superficial flashcards, and reinforces learning with gamified XP, leveling, streaks, and Anki-style SM-2 spaced repetition.*

2. System Architecture & Tech Stack

Layer	Technology	Key Responsibilities
Frontend UI	HTML5, Tailwind CSS, Lucide Icons, Canvas Confetti	Responsive SPA (Desktop sidebar + Mobile bottom nav), 3D flip card animations, XP celebrations, Dark/Light modes.
Backend API	FastAPI, Uvicorn, Python 3.11	REST API endpoints, file uploads handling, business logic orchestration, spaced repetition computations.
AI Services	Sarvam AI Python SDK (sarvamai)	**Doc AI Digitise:** Handwriting and textbook OCR. **sarvam-105b:** Conceptual flashcard and quiz generation.
Persistence	SQLite (data/app.db), Local File Storage	Zero-configuration local database saving notes, cards, quizzes, streaks, XP, logs, and uploaded image buffers.

3. Sarvam AI Integration Deep Dive

A. Document Intelligence & Handwriting OCR (client.doc_ai.digitise)

Users can upload photos of handwritten notebooks, lecture whiteboard slides, or textbook pages. The system dispatches the image to Sarvam's asynchronous Doc AI digitise pipeline. The engine identifies document layout structures including paragraphs, headings, lists, tables, and formulas. The application polls for completion, receives structured JSON coordinates and normalized bounding boxes (bbox_norm), and presents the extracted text in an editable verification workspace alongside the uploaded image so users can review before generating cards.

B. Multilingual Cognitive Material Generation (sarvam-105b-conversations)

Once verified text is ready, Sarvam's 105B parameter conversational model processes the text through a specialized educational prompt. Rather than producing rote trivia, it creates conceptual question-and-answer flashcards and 4-option multiple-choice quizzes with detailed rationales. **Multilingual Adaptability:** Sarvam AI natively understands English, 11+ Indian languages (Hindi, Tamil, Telugu, Kannada, Bengali, Marathi, etc.), and code-mixed bilingual inputs (such as Hinglish). When study notes contain non-English or mixed text, the generated questions and explanations naturally adopt that same language style.

4. Core Features & Learning Mechanics

- **Daily Home Dashboard:** Presents today's study plan at a glance: flashcards due, new cards, daily goal progress, streak flames, weekly study activity chart, and weak topics diagnostics.
- **SuperMemo (SM-2) Spaced Repetition:** Implements the proven SM-2 memory retention algorithm. Each flashcard tracks repetition count, ease factor (1.3 to 3.0), and interval days. Rating cards as Again (<1d), Hard (~1-2d), Good (~4-6d), or Easy (~8-12d) dynamically recalculates optimal review intervals.
- **Interactive 3D Flashcards:** Full 3D card flip animation on click or Spacebar. Back of card displays the verified answer, deep explanation, subject tags, and source note reference. Rating buttons are mapped to keyboard shortcuts 1, 2, 3, 4.
- **Gamified Interactive Quizzes:** Dynamic multiple-choice questions with instantaneous visual state change upon selection (green for correct, red for incorrect), followed by clear conceptual explanations and XP awards.
- **Gamification & Progression System:** Features XP accumulation, level progression (Level 1 Novice → Level 5 Mind Architect), streak counters, and an unlockable achievement badges shelf (e.g. First Step, Sharp Mind, Flame Keeper, Visionary).
- **Local Persistence & Privacy:** All notes, flashcards, quizzes, study logs, and images are stored locally in SQLite (data/app.db) and data/uploads/, ensuring context is never lost across sessions without requiring mandatory logins.

5. Database Schema & Storage Model

The application utilizes SQLite with 5 primary tables: **notes** (source text, image paths, OCR blocks), **flashcards** (note relation, Q&A;, ease factor, repetition count, due dates), **quizzes** (question, JSON options array, correct index, historical accuracy), **user_stats** (total XP, current level, daily streaks, unlocked badge IDs, weekly counts), and **study_logs** (audit history of every review and quiz attempt for telemetry). Users can export the complete database as a portable JSON backup at any time from the Settings view.

6. Setup & Execution Instructions

- **Step 1 — Environment Configuration:** Place your unified Sarvam API key in .env as SARVAM_API_KEY=your_key_here (already protected via .gitignore).
- **Step 2 — Launch Application:** Run python run.py in the terminal. The server starts instantly on http://127.0.0.1:8000.
- **Step 3 — Access Web UI:** Open http://127.0.0.1:8000 in your browser. You can start reviewing pre-seeded demo decks or click **Add Knowledge** to digitize notes.